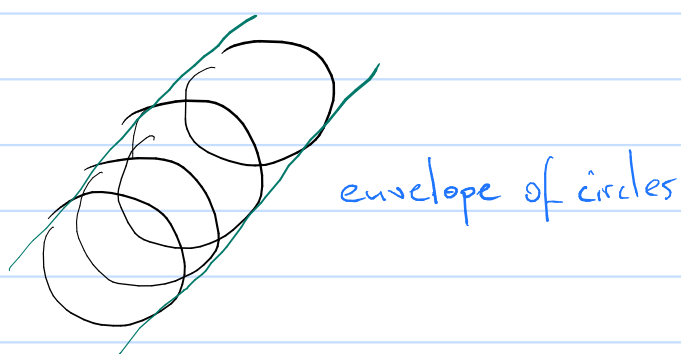


Envelopes of curves



Let $F: \mathbb{R}^2 \times \mathbb{R} \rightarrow \mathbb{R}$ be a smooth function. Then the incidence surface $S = \{(x, y, t) \in \mathbb{R}^2 \times \mathbb{R} : F(x, y, t) = 0\} \subset \mathbb{R}^2 \times \mathbb{R}$

is a smooth surface. Let $\pi: \mathbb{R}^2 \times \mathbb{R} \rightarrow \mathbb{R}$ denote the projection onto the first factor. Let $t \in \mathbb{R}$ and let C_t be the projection of the t -level curve in S . We consider the set $\{C_t : t \in \mathbb{R}\}$ a family of curves. We call the set of critical values of $\pi|_S$ the envelope of the family of curves.

Geometric interpretation: Let $(x_0, y_0, t_0) \in S$. Then

$$T_{(x_0, y_0, t_0)} S = \{(u, v, w) \in \mathbb{R}^2 \times \mathbb{R} : F_x u + F_y v + F_t w = 0\},$$

and $d\pi(u, v, w) = (u, v)$. Now, $\pi|_S$ is singular at (x_0, y_0, t_0) iff $\text{Ker } d\pi|_{(x_0, y_0, t_0)} \neq 0$, i.e. iff $(0, 0, w) \in S$ for $w \neq 0$. Hence, $\pi|_S$ is singular at (x_0, y_0, t_0) iff

$$F_t(x_0, y_0, t_0) = 0.$$

Intuitively, the curves C_{t_0} and $C_{t_0 + \Delta t}$ intersect at (x_0, y_0) at first order.

In particular, the envelope of $\{C_t\}$ is the curve

$$E := \{(x, y) \in \mathbb{R}^2 : \exists t \in \mathbb{R} \text{ s.t. } F(x, y, t) = 0 \text{ and } F_t(x, y, t) = 0\}.$$

Note that, on E , the tangent to S becomes

$$uF_x + vF_y = 0$$

for $(u, v, t) \in T_{(x, y, t)} S$. Taking a local parameterisation $x = x(t)$, $y = y(t)$, and differentiating $F(x(t), y(t), t)$, we see that this is also the equation for the tangent to the envelope at (x_0, y_0, t) . Hence, an alternative interpretation of the envelope is the curve on $\pi(S)$ of points that are on exactly one curve C_t (locally in t) and share their tangents with C_t .

$$f(x, y) = 0 \text{ has tangents}$$

$$\frac{\partial f}{\partial x} (x - x^0) + \frac{\partial f}{\partial y} (y - y^0) = 0$$

Legendre Transform

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be smooth and convex. Note that its graph

$$E = \{(x, y) \in \mathbb{R}^2 : y - f(x) = 0\}$$

is the envelope for its family of tangent lines. These are given by

$$\{y = x f'(x_0) + \underbrace{y_0 - x_0 f'(x_0)}_{f(x_0)}\}$$

for $(x_0, y_0) \in E$. This is parameterised by $x_0 \in \mathbb{R}$. We rewrite this family as

$$\{y = p(x_0)x + b(x_0)\}_{x_0},$$

with $p(x_0) = f'(x_0)$. Since f is convex, we get

$$x_0 = (f')^{-1}(p),$$

and hence

$$b = f(x_0) - p(x_0)x_0 = f((f')^{-1}(p)) - p(f')^{-1}(p) =: -f^*(p),$$

the **Legendre transform** of f . In particular, this family of lines is also parameterised by p :

$$\{y = px - f^*(p)\}_p.$$

Let $F(x, y, p) = y + f^*(p) - px$. As an envelope, the original graph is the solution to

$$\begin{cases} F(x, y, p) = 0 \\ F_p(x, y, p) = (f^*)'(p) - x = 0. \end{cases}$$

We get

$$x = (f^*)'(p) \Rightarrow p = ((f^*)')^{-1}(x)$$

and hence

$$y = x((f^*)')^{-1}(x) - f^*((f^*)')^{-1}(x)),$$

which is just the Legendre transform of f^* . Hence,

$$f = f^{**}.$$

